

# Trading Model Features Documentation

This document provides a complete reference for all features used in the hybrid trading model, including their mathematical formulas, data sources, and trading rationale.

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## Meta Features

### Basic Identifiers

| Feature    | Type     | Description                                   |
|------------|----------|-----------------------------------------------|
| ticker     | string   | Symbol being traded (SPX, SPY, QQQ, /ES, /NQ) |
| date       | string   | Trading date in YYYYMMDD format               |
| time       | string   | Time of day in HH:MM format                   |
| timestamp  | datetime | Full timestamp of the observation             |
| spot_price | float    | Current market price of the underlying        |

# Target Labels

## Target Classification

**Formula:**

```
target = {
  1    if max_up_move ≥ 0.4% AND max_up_move > max_down_move    (LONG)
 -1    if max_down_move ≥ 0.4% AND max_down_move > max_up_move    (SHORT)
  0    otherwise    (HOLD)
}
```

**where:**

```
max_up_move = (max_price_next_120min - current_price) / current_price
max_down_move = (current_price - min_price_next_120min) /
current_price
```

**Threshold:** 0.4% (TARGET\_MOVE\_THRESHOLD)

**Lookahead Window:** 120 minutes

## Timing Features

| Feature                     | Formula                                              | Description                                        |
|-----------------------------|------------------------------------------------------|----------------------------------------------------|
| <code>time_to_target</code> | Minutes until 0.4% target hit                        | Time to reach profit target (0 if not hit)         |
| <code>time_to_stop</code>   | Minutes until 0.4% stop hit                          | Time to hit opposite-direction stop (0 if not hit) |
| <code>max_move</code>       | $\max(\text{max\_up\_move}, \text{max\_down\_move})$ | Maximum favorable price movement percentage        |

**Purpose:** These provide the model with information about how long profitable moves typically take to develop, helping optimize entry timing.

# ODTE Greek Exposures

ODTE (zero days to expiration) options dominate intraday flow and create immediate hedging pressure on market makers.

## Net Greek Values

All net Greeks are calculated by summing the exposure across all strikes:

$$\text{net\_greek} = \sum (\text{greek\_exposure}[\text{strike}]) \text{ for all strikes}$$

| Feature   | Source Greek | Formula               | Trading Significance                                                                                                                   |
|-----------|--------------|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| net gamma | totalgamma   | $\sum(\text{all}[i])$ | <b>Volatility dampening/amplification.</b> Positive = dealers long gamma (stabilizing), negative = dealers short gamma (destabilizing) |
| net vanna | totalvanna   | $\sum(\text{all}[i])$ | <b>Vol-spot correlation.</b> Positive vanna means rising IV → rising spot (bullish feedback loop)                                      |
| net charm | totalcharm   | $\sum(\text{all}[i])$ | <b>Time decay directional pressure.</b> Positive charm = theta decay pushes price up                                                   |
| net dgex  | totaldgex    | $\sum(\text{all}[i])$ | <b>Dealer gamma exposure.</b> Shows where dealers need to hedge most aggressively                                                      |
| net zomma | totalzomma   | $\sum(\text{all}[i])$ | <b>Gamma stability.</b> Positive = gamma increases with vol (stabilizing effect)                                                       |
| net delta | totaldelta   | $\sum(\text{all}[i])$ | <b>Directional exposure.</b> Net directional pressure from all options                                                                 |
| net vega  | totalvega    | $\sum(\text{all}[i])$ | <b>Volatility exposure.</b> Sensitivity to changes in implied volatility                                                               |
| net vomma | totalvomma   | $\sum(\text{all}[i])$ | <b>Vol-of-vol exposure.</b> Second-order sensitivity (convexity) to volatility changes                                                 |

## Greek Regime Classification

Formula:

```
gamma_regime = {
    2 if net_gamma > 0.1      # Long gamma (dealers stabilize)
    0 if net_gamma < -0.1    # Short gamma (dealers amplify)
    1 otherwise              # Neutral
}
```

Binary Greek Signals:

| Feature       | Formula                     | Interpretation                   |
|---------------|-----------------------------|----------------------------------|
| vanna_bullish | 1 if net_vanna > 0.1 else 0 | Vol-spot feedback is bullish     |
| charm_bullish | 1 if net_charm > 0.1 else 0 | Time decay favors upside         |
| dgex_sticky   | 1 if net_dgex > 0.1 else 0  | Dealers pin price (high hedging) |

| Feature           | Formula                        | Interpretation                          |
|-------------------|--------------------------------|-----------------------------------------|
| zomma_stabilizing | 1 if net_zomma > 0.1 else 0    | Gamma becomes more stabilizing with vol |
| vega_elevated     | 1 if abs(net_vega)> 0.1 else 0 | High sensitivity to volatility changes  |

## ODTE Key Levels

Key levels are strikes where Greek exposure is concentrated, creating natural support/resistance.

### Strike Identification

#### Max/Min Greek Strikes:

```

max_gamma_strike = levels[argmax(totalgamma.all)]
min_gamma_strike = levels[argmin(totalgamma.all)]
max_dgex_strike = levels[argmax(totaldgex.all)]
min_dgex_strike = levels[argmin(totaldgex.all)]
min_vanna_strike = levels[argmin(totalvanna.all)]
max_vega_strike = levels[argmax(totalvega.all)]
min_vega_strike = levels[argmin(totalvega.all)]
max_vomma_strike = levels[argmax(totalvomma.all)]
min_vomma_strike = levels[argmin(totalvomma.all)]

```

#### Zero Crossings:

- zero\_gamma : Strike where net gamma changes sign
- zero\_delta : Strike where net delta changes sign

### Distance Features

All distances are normalized by current price:

```

dist_to_level = (spot_price - level) / spot_price

```

| Feature           | Formula                   | Trading Use                                           |
|-------------------|---------------------------|-------------------------------------------------------|
| dist_to_max_gamma | (spot - max_gamma) / spot | Distance to strongest gamma wall (resistance/support) |

| Feature                         | Formula                                            | Trading Use                              |
|---------------------------------|----------------------------------------------------|------------------------------------------|
| <code>dist_to_min_gamma</code>  | $(\text{spot} - \text{min\_gamma}) / \text{spot}$  | Distance to gamma hole (volatility zone) |
| <code>dist_to_min_vanna</code>  | $(\text{spot} - \text{min\_vanna}) / \text{spot}$  | Distance to vanna flip point             |
| <code>dist_to_zero_gamma</code> | $(\text{spot} - \text{zero\_gamma}) / \text{spot}$ | Distance to gamma transition level       |
| <code>dist_to_max_dgex</code>   | $(\text{spot} - \text{max\_dgex}) / \text{spot}$   | Distance to peak dealer hedging          |
| <code>dist_to_min_dgex</code>   | $(\text{spot} - \text{min\_dgex}) / \text{spot}$   | Distance to dealer hedging hole          |
| <code>dist_to_max_vega</code>   | $(\text{spot} - \text{max\_vega}) / \text{spot}$   | Distance to peak volatility exposure     |
| <code>dist_to_min_vega</code>   | $(\text{spot} - \text{min\_vega}) / \text{spot}$   | Distance to volatility hole              |
| <code>dist_to_max_vomma</code>  | $(\text{spot} - \text{max\_vomma}) / \text{spot}$  | Distance to peak convexity               |
| <code>dist_to_min_vomma</code>  | $(\text{spot} - \text{min\_vomma}) / \text{spot}$  | Distance to convexity hole               |

## Proximity Flags

### Formula:

```
near_level = 1 if |spot - level| / spot ≤ 0.04% else 0
```

| Feature                      | Condition                  | Significance                                        |
|------------------------------|----------------------------|-----------------------------------------------------|
| <code>near_max_gamma</code>  | Within 0.04% of max gamma  | At gamma wall - expect rejection or breakout        |
| <code>near_min_gamma</code>  | Within 0.04% of min gamma  | In gamma hole - expect acceleration                 |
| <code>near_min_vanna</code>  | Within 0.04% of min vanna  | At vanna inflection - vol/spot relationship changes |
| <code>near_zero_gamma</code> | Within 0.04% of zero gamma | At regime transition point                          |

**Trading Logic:** When price is "near" a key level, the market often shows distinct behavior (consolidation, reversal, or breakout).

# Weekly Greek Exposures

Weekly options provide intermediate-term context (1-4 weeks to expiration) and show institutional positioning.

All weekly features use the same formulas as ODTE Greeks but with `wk_` prefix and are normalized to  $[0, 1]$  range where applicable.

## Net Weekly Greeks

| Feature                   | Source            | Normalized Formula      |
|---------------------------|-------------------|-------------------------|
| <code>wk_net_gamma</code> | weekly totalgamma | $\Sigma(\text{all}[i])$ |
| <code>wk_net_vanna</code> | weekly totalvanna | $\Sigma(\text{all}[i])$ |
| <code>wk_net_charm</code> | weekly totalcharm | $\Sigma(\text{all}[i])$ |
| <code>wk_net_dgex</code>  | weekly totaldgex  | $\Sigma(\text{all}[i])$ |
| <code>wk_net_zomma</code> | weekly totalzomma | $\Sigma(\text{all}[i])$ |
| <code>wk_net_delta</code> | weekly totaldelta | $\Sigma(\text{all}[i])$ |
| <code>wk_net_vega</code>  | weekly totalvega  | $\Sigma(\text{all}[i])$ |
| <code>wk_net_vomma</code> | weekly totalvomma | $\Sigma(\text{all}[i])$ |

## Weekly Key Levels

| Feature                           | Formula                                               | Purpose                         |
|-----------------------------------|-------------------------------------------------------|---------------------------------|
| <code>wk_dist_to_max_gamma</code> | $(\text{spot} - \text{wk\_max\_gamma}) / \text{spot}$ | Weekly gamma wall location      |
| <code>wk_dist_to_min_gamma</code> | $(\text{spot} - \text{wk\_min\_gamma}) / \text{spot}$ | Weekly gamma hole location      |
| <code>wk_dist_to_max_dgex</code>  | $(\text{spot} - \text{wk\_max\_dgex}) / \text{spot}$  | Weekly dealer pin level         |
| <code>wk_dist_to_min_dgex</code>  | $(\text{spot} - \text{wk\_min\_dgex}) / \text{spot}$  | Weekly dealer hedging hole      |
| <code>wk_dist_to_max_vega</code>  | $(\text{spot} - \text{wk\_max\_vega}) / \text{spot}$  | Weekly volatility exposure peak |
| <code>wk_dist_to_min_vega</code>  | $(\text{spot} - \text{wk\_min\_vega}) / \text{spot}$  | Weekly volatility hole          |

## Weekly Regime

| Feature | Formula | Range |
|---------|---------|-------|
|---------|---------|-------|

| Feature              | Formula                        | Range                                     |
|----------------------|--------------------------------|-------------------------------------------|
| wk_gamma_regime      | wk_gamma_regime / 2.0          | [0, 1] where 0=short, 0.5=neutral, 1=long |
| wk_vanna_bullish     | 1 if wk_net_vanna > 0.1 else 0 | Binary                                    |
| wk_dgex_sticky       | 1 if wk_net_dgex > 0.1 else 0  | Binary                                    |
| wk_zomma_stabilizing | 1 if wk_net_zomma > 0.1 else 0 | Binary                                    |

**Purpose:** Weekly Greeks show where institutions are positioned for multi-day moves, while ODTE shows intraday hedging flow.

## Cross-Expiry Divergence

Divergence between ODTE and weekly Greeks indicates flow disagreement and potential regime changes.

### Sign Divergence Formula

```
def sign_divergence(a: float, b: float) -> float:
    if |a| < 0.01 or |b| < 0.01:
        return 0.5 # One side is neutral
    if sign(a) != sign(b):
        return 1.0 # Divergence (conflicting signals)
    return 0.0 # Agreement
```

| Feature          | Comparison                        | Trading Interpretation                              |
|------------------|-----------------------------------|-----------------------------------------------------|
| gamma_0dte_vs_wk | sign_div(net_gamma, wk_net_gamma) | 1.0 = intraday vs multi-day regime conflict         |
| vanna_0dte_vs_wk | sign_div(net_vanna, wk_net_vanna) | 1.0 = vol-spot feedback disagreement                |
| dgex_0dte_vs_wk  | sign_div(net_dgex, wk_net_dgex)   | 1.0 = intraday vs multi-day dealer hedging conflict |
| delta_0dte_vs_wk | sign_div(net_delta, wk_net_delta) | 1.0 = directional flow disagreement                 |
| vega_0dte_vs_wk  | sign_div(net_vega, wk_net_vega)   | 1.0 = volatility exposure mismatch                  |

| Feature                       | Comparison                                     | Trading Interpretation   |
|-------------------------------|------------------------------------------------|--------------------------|
| <code>vomma_0dte_vs_wk</code> | <code>sign_div(net_vomma, wk_net_vomma)</code> | 1.0 = convexity mismatch |

**Trading Edge:** High divergence (1.0) often precedes reversals or regime changes as one timeframe "wins" the tug-of-war.

## Initial Balance (IB) Features

Initial Balance is the price range established in the first hour of trading (9:30-10:30 AM ET). It's a critical framework for intraday trading.

### IB Range Definition

```
ib_high = max(price) for t ∈ [9:30, 10:30]
ib_low = min(price) for t ∈ [9:30, 10:30]
ib_range = ib_high - ib_low
```

### Price Relative to IB

| Feature                       | Formula                                            | Interpretation                            |
|-------------------------------|----------------------------------------------------|-------------------------------------------|
| <code>price_vs_ib_high</code> | $(\text{spot} - \text{ib\_high}) / \text{spot}$    | Positive = above IB (bullish breakout)    |
| <code>price_vs_ib_low</code>  | $(\text{spot} - \text{ib\_low}) / \text{spot}$     | Negative = below IB (bearish breakdown)   |
| <code>ib_range_pct</code>     | $(\text{ib\_high} - \text{ib\_low}) / \text{spot}$ | IB volatility (wider = more volatile day) |

### IB Proximity Flags

| Feature                   | Condition                                                   | Trading Signal                          |
|---------------------------|-------------------------------------------------------------|-----------------------------------------|
| <code>near_ib_high</code> | $ \text{spot} - \text{ib\_high}  / \text{spot} \leq 0.04\%$ | Testing IB high - breakout or rejection |
| <code>near_ib_low</code>  | $ \text{spot} - \text{ib\_low}  / \text{spot} \leq 0.04\%$  | Testing IB low - breakdown or bounce    |

### IB Position Flags

| Feature               | Condition                      | Trading Context                  |
|-----------------------|--------------------------------|----------------------------------|
| <code>above_ib</code> | <code>spot &gt; ib_high</code> | Bullish - IB high is now support |

| Feature                  | Condition                            | Trading Context                    |
|--------------------------|--------------------------------------|------------------------------------|
| <code>below_ib</code>    | <code>spot &lt; ib_low</code>        | Bearish - IB low is now resistance |
| <code>in_ib_range</code> | <code>ib_low ≤ spot ≤ ib_high</code> | Consolidation - expect breakout    |

Trading Framework:

- 68% of days stay within IB ± 1 extension
- Breakouts above/below IB tend to continue
- Returns to IB after breakout often fail (trap)

## Fibonacci Extensions

Fibonacci levels project probable price targets based on IB range using classical ratios.

### Calculation

```
ib_range = ib_high - ib_low

# Upside extensions (from IB low)
fib_127_up = ib_low + (ib_range × 1.272)
fib_161_up = ib_low + (ib_range × 1.618)
fib_200_up = ib_low + (ib_range × 2.000)

# Downside extensions (from IB low)
fib_127_dn = ib_low + (ib_range × -0.272)
fib_161_dn = ib_low + (ib_range × -0.618)
fib_200_dn = ib_low + (ib_range × -1.000)
```

## Distance Features

| Feature                      | Formula                                 | Usage                                             |
|------------------------------|-----------------------------------------|---------------------------------------------------|
| <code>dist_fib_127_up</code> | <code>(spot - fib_127_up) / spot</code> | Distance to 1.272 extension (first target)        |
| <code>dist_fib_161_up</code> | <code>(spot - fib_161_up) / spot</code> | Distance to 1.618 extension (golden ratio target) |

Trading Application:

- 1.272 extension: ~50% of breakouts reach this level
- 1.618 extension: ~30% of breakouts reach this level (strong move)

- 2.0 extension: ~10% of breakouts (rare, momentum day)

# Implied Volatility (IV) Features

IV features are derived from ATM put options using Fourier filtering to remove noise.

## Raw IV

**Source:** ATM put implied volatility from options chain

**Formula:**

```
atm_iv = IV(ATM_put) / 100 # Normalized to decimal (e.g., 0.15 = 15%)
```

## IV Statistical Context

**Daily Statistics:**

```
iv_mean = mean(atm_iv[t]) for all t in trading_day
iv_std = std(atm_iv[t]) for all t in trading_day
iv_min = min(atm_iv[t]) for all t in trading_day
iv_max = max(atm_iv[t]) for all t in trading_day
```

| Feature       | Formula                                                                       | Interpretation                                                         |
|---------------|-------------------------------------------------------------------------------|------------------------------------------------------------------------|
| iv_zscore     | $\text{clip}((\text{atm\_iv} - \text{iv\_mean}) / \text{iv\_std}, -3, 3) / 3$ | Normalized to [-1, 1]; 1.0 = IV is $3\sigma$ above mean (extreme fear) |
| iv_percentile | $(\text{atm\_iv} - \text{iv\_min}) / (\text{iv\_max} - \text{iv\_min})$       | Where current IV ranks in today's range [0, 1]                         |

**Trading Logic:**

- High IV Z-score (>0.5): Options expensive, consider selling premium or fading moves
- Low IV Z-score (<-0.5): Options cheap, consider buying options or expecting expansion
- IV percentile > 0.8: IV near daily highs, likely to contract (mean reversion)

## VIX Features

VIX (CBOE Volatility Index) provides market-wide fear/greed context independent of underlying-specific Greeks.

## VIX Data Sources

- 1. **VIX Spot:** From IB minute bars (VIX index itself)
- 2. **VIX Gamma:** From VIX weekly options exposure (dealers' VIX hedging needs)

## VIX Metrics

| Feature   | Formula                                     | Range   | Interpretation                                    |
|-----------|---------------------------------------------|---------|---------------------------------------------------|
| vix_spot  | VIX_index_price / 50                        | [0, 1+] | Normalized VIX level (0.3 = 15 VIX, 0.5 = 25 VIX) |
| vix_gamma | $\Sigma(\text{VIX\_weekly\_totalgamma}[i])$ | Raw     | Net gamma in VIX options (hedging demand)         |

## VIX Regime Classification

Formula:

```
vix_regime = {
    2 if vix_spot > 25      # High fear
    1 if vix_spot > 18      # Elevated anxiety
    0 otherwise             # Low/complacent
}

vix_regime_normalized = vix_regime / 2.0 # Scale to [0, 1]
```

| VIX Level | Regime       | Typical Market Behavior                   |
|-----------|--------------|-------------------------------------------|
| < 18      | Low (0)      | Complacent, low hedging, trending markets |
| 18-25     | Elevated (1) | Moderate uncertainty, choppy              |
| > 25      | High (2)     | Fear, aggressive hedging, volatile        |

Trading Application:

- VIX > 25 + positive gamma = expect mean reversion (dealers stabilize)
- VIX > 25 + negative gamma = expect continued volatility (dealers amplify)
- VIX < 18 = "vol crush" regime, fade extended moves

# Market Context

## Technical Indicators

### RSI (Relative Strength Index)

#### Formula:

```
def rsi(prices, period=14):
    deltas = diff(prices)
    gains = where(deltas > 0, deltas, 0)
    losses = where(deltas < 0, -deltas, 0)

    avg_gain = mean(gains[-period:])
    avg_loss = mean(losses[-period:])

    rs = avg_gain / avg_loss
    rsi = 100 - (100 / (1 + rs))
    return rsi / 100 # Normalized to [0, 1]
```

#### Interpretation:

- RSI > 0.7(70): Overbought, potential reversal
- RSI < 0.3(30): Oversold, potential bounce
- RSI ~ 0.5(50): Neutral, no momentum bias

**Why #3 Feature:** RSI provides critical momentum confirmation alongside Greek dynamics. The model uses it to filter out false Greek signals during overbought/oversold extremes.

### Volume Relative

#### Formula:

```
avg_volume_per_bar = total_daily_volume / num_bars
vol_relative = current_bar_volume / avg_volume_per_bar
vol_relative_normalized = min(vol_relative, 5.0) / 5.0 # Capped at 5x av
```

#### Interpretation:

- vol\_relative > 2.0: High volume bar, significant price action
- vol\_relative < 0.5: Low volume bar, lack of conviction
- Volume spikes confirm breakouts, lack of volume suggests false moves

# Time Features

| Feature           | Formula       | Purpose                       |
|-------------------|---------------|-------------------------------|
| hour_normalized   | hour / 24.0   | Time of day effect [0, 1]     |
| minute_normalized | minute / 60.0 | Intra-hour positioning [0, 1] |

## Intraday Patterns:

- 9:30-10:30: Initial balance formation, high volatility
  - 10:30-11:30: Trend development or IB test
  - 11:30-14:00: Lunch lull, low volume/volatility
  - 14:00-15:00: Afternoon continuation or reversal
  - 15:00-16:00: Closing drive, profit-taking, stops triggered
- 

# RBF Confluences

Radial Basis Function (RBF) confluences measure the overlap between structural levels (IB, Fibonacci) and Greek levels using a Gaussian kernel.

## Formula:

```
def rbf_confluence(level_a, level_b, spot, sigma=0.05):  
    dist = abs(level_a - level_b) / spot  
    return exp(-dist**2 / (2 * sigma**2)) # Returns [0, 1]
```

## Interpretation:

- 1.0: Levels are identical (Strong Confluence)
- 0.0: Levels are far apart (No Confluence)
- High confluence indicates a "super level" where multiple factors align.

## Key Confluences:

| Feature                         | Combination             | Significance                                   |
|---------------------------------|-------------------------|------------------------------------------------|
| confluence_ib_high_max_gamma    | IB High + Max Gamma     | Structural resistance backed by gamma wall     |
| confluence_ib_low_min_gamma     | IB Low + Min Gamma      | Structural support at volatility trigger point |
| confluence_fib161_bull_max_vega | Fib 1.618 Up + Max Vega | Bullish target aligns with high vol exposure   |

| Feature                         | Combination               | Significance                                   |
|---------------------------------|---------------------------|------------------------------------------------|
| confluence_fib127_bear_min_dgex | Fib 1.272 Down + Min DGEX | Bearish target aligns with dealer hedging hole |

## Engineered Features

These features combine multiple inputs to capture complex market dynamics.

### Greek Ratios

Ratios reveal the relative strength of different Greek forces:

| Feature           | Formula                                                | Interpretation                                                            |
|-------------------|--------------------------------------------------------|---------------------------------------------------------------------------|
| gamma_vanna_ratio | $\text{net\_gamma} / ( \text{net\_vanna}  + \epsilon)$ | Gamma vs vanna dominance; high = gamma effects override vol-spot feedback |
| dgex_gamma_ratio  | $\text{net\_dgex} / ( \text{net\_gamma}  + \epsilon)$  | Dealer hedging intensity vs gamma regime                                  |
| charm_vanna_ratio | $\text{net\_charm} / ( \text{net\_vanna}  + \epsilon)$ | Time decay pressure vs vol-spot feedback                                  |
| delta_gamma_ratio | $\text{net\_delta} / ( \text{net\_gamma}  + \epsilon)$ | Directional flow vs gamma regime                                          |
| vega_gamma_ratio  | $\text{net\_vega} / ( \text{net\_gamma}  + \epsilon)$  | Volatility exposure vs gamma stabilizing/destabilizing                    |
| vomma_vega_ratio  | $\text{net\_vomma} / ( \text{net\_vega}  + \epsilon)$  | Convexity vs linear volatility exposure                                   |

**Note:**  $\epsilon = 1\text{e-}6$  prevents division by zero

#### Trading Use:

- High gamma\_vanna\_ratio: Price stability dominates (gamma pinning)
- High dgex\_gamma\_ratio: Aggressive dealer hedging, expect sharp moves
- Negative charm\_vanna\_ratio with high magnitude: Complex feedback loops

### Temporal Deltas

Changes between consecutive observations reveal flow acceleration/deceleration:

| Feature | Formula | Interpretation |
|---------|---------|----------------|
|---------|---------|----------------|

| Feature      | Formula                           | Interpretation                            |
|--------------|-----------------------------------|-------------------------------------------|
| gamma_change | net_gamma(t) - net_gamma(t-1)     | Gamma accumulation/decay rate             |
| vanna_change | net_vanna(t) - net_vanna(t-1)     | Vol-spot feedback strengthening/weakening |
| dgex_change  | net_dgex(t) - net_dgex(t-1)       | Dealer hedging pressure change            |
| delta_change | net_delta(t) - net_delta(t-1)     | Directional flow acceleration             |
| spot_change  | (spot(t) - spot(t-1)) / spot(t-1) | Price momentum                            |
| vega_change  | net_vega(t) - net_vega(t-1)       | Volatility exposure change                |
| vomma_change | net_vomma(t) - net_vomma(t-1)     | Convexity change                          |

Trading Logic:

- Positive gamma\_change + rising price = gamma wall building above (resistance)
- Negative dgex\_change = dealers reducing hedging (expect mean reversion)
- Large spot\_change with small delta\_change = move driven by technical factors, not flow

Cross-Features

Complex interactions between price action and Greek dynamics:

Gamma Momentum

Formula:

```
gamma_momentum = (net_gamma(t) - net_gamma(t-1)) * sign(net_gamma(t))
```

Interpretation:

- Positive gamma\_momentum (>0): Long gamma regime strengthening → expect stabilization
- Negative gamma\_momentum (<0): Short gamma regime deepening → expect volatility expansion

Price vs DGEX Magnet

Formula:

```
price_vs_dgex_magnet = (
    (spot(t) - spot(t-1)) / spot(t-1) # Price change direction
    *
    sign((spot(t) - max_dgex_strike(t)) / spot(t)) # Position relative t
)
```

Interpretation:

- Positive: Price moving toward max DGEX (magnetic pull)
- Negative: Price moving away from max DGEX (breaking free or reversal)
- Magnitude: Speed of attraction/repulsion

**Trading Edge:** Max DGEX strikes act as "magnets" due to dealer hedging. When price approaches max DGEX with positive price\_vs\_dgex\_magnet, expect consolidation. When negative, expect breakout continuation.

Feature Summary Table

| Category       | Count | Key Features                                                         | Primary Use            |
|----------------|-------|----------------------------------------------------------------------|------------------------|
| Meta           | 5     | ticker, date, time, timestamp, spot_price                            | Identification         |
| Target         | 4     | target, time_to_target, time_to_stop, max_move                       | Model output           |
| ODTE Greeks    | 17    | <b>net_charm (#1)</b> , net_vanna, net_dgex, <b>dgex_sticky (#5)</b> | Intraday flow          |
| Weekly Greeks  | 14    | wk_net_gamma, wk_net_vanna, wk_dgex, distances                       | Multi-day context      |
| Cross-Expiry   | 4     | gamma_0dte_vs_wk, vanna_0dte_vs_wk, etc.                             | Regime detection       |
| IB Features    | 8     | price_vs_ib_high, near_ib_high, in_ib_range                          | Intraday framework     |
| Fibonacci      | 2     | dist_fib_127_up, dist_fib_161_up                                     | Target projection      |
| IV Features    | 4     | atm_iv, iv_zscore, iv_percentile                                     | Vol environment        |
| VIX Features   | 3     | vix_spot, vix_gamma, vix_regime                                      | Market-wide sentiment  |
| Market Context | 3     | <b>rsi (#3)</b> , vol_relative, time features                        | Technical confirmation |
| Engineered     | 10    | ratios, deltas, gamma_momentum, dgex_magnet                          | Complex dynamics       |
| Key Levels     | -     | <b>dist_to_min_gamma (#2)</b> , <b>dist_to_min_vanna (#4)</b>        | Critical positioning   |

**Total Features:** ~74 numerical features (excluding meta fields)

Data Flow Pipeline

## 1. Raw Data Collection

- └ Greek JSONs (0DTE & Weekly) → Net exposures, key strikes
- └ IB Minute Bars → Price series, volume, IB range
- └ Fourier IV Data → ATM IV time series
- └ VIX Data → VIX spot, VIX gamma

## 2. Feature Engineering

- └ Greek Processing → Net values, distances, flags
- └ Level Detection → Max/min strikes, zero crossings
- └ Temporal Joins → Match Greek snapshots to price bars
- └ Statistical Context → RSI, IV percentiles, volume

## 3. Cross-Feature Generation

- └ Greek Ratios → Relative strength of forces
- └ Temporal Deltas → Rate of change
- └ Complex Interactions → Gamma momentum, DGEX magnet

## 4. Target Calculation

- └ Lookahead Analysis → Price moves over next 120 minutes

## 5. Dataset Output

- └ CSV with 93,387 samples × 74+ features
- 

# Feature Importance Insights

From the latest model training (medium architecture), the top 5 most important features are:

### 1. **net\_charm** (0.2154 / 21.5%): Time decay directional pressure - **dominant feature by far**

- Theta decay creates directional bias as options approach expiration
- Positive charm = upward pressure, negative charm = downward pressure
- This is the #1 signal the model uses to predict intraday moves

### 2. **dist\_to\_min\_gamma** (0.0562 / 5.6%): Distance to gamma hole (volatility acceleration zone)

- Gamma holes are strikes with minimal dealer hedging
- Price approaching gamma holes = expect acceleration/volatility
- Critical for predicting breakout magnitude

### 3. **rsi** (0.0482 / 4.8%): Relative Strength Index momentum indicator

- Technical confirmation of Greek signals
- Filters false signals during extreme overbought/oversold conditions

- Shows model benefits from combining Greeks + technicals

4. **dist\_to\_min\_vanna** (0.0316 / 3.2%): Distance to vanna inflection point

- Vanna controls vol-spot correlation
- At vanna flip points, volatility-price relationship inverts
- Key for understanding vol expansion/contraction dynamics

5. **dgex\_sticky** (0.0297 / 3.0%): Binary flag for high dealer gamma exposure

- Indicates dealer "pinning" behavior near strikes
- When active, expect price consolidation/resistance to movement
- Important for avoiding low-probability breakout attempts

## Key Insights

**Charm Dominance:** The model places **7x more weight** on charm than any other feature (21.5% vs 5.6% for #2). This suggests that **time decay effects are the primary edge** in intraday trading - understanding how theta creates directional pressure as options approach expiration is more important than raw gamma/vanna exposure.

**Level-Based Thinking:** Three of the top 5 features (#2, #4, #5) are about **positioning relative to critical levels** rather than absolute Greek values. This indicates the model has learned that **where you are matters more than what the aggregate flow is**.

**Technical + Greeks Synergy:** RSI at #3 shows the model **combines Greeks with momentum** - it's not purely Greeks-based. Technical overbought/oversold conditions help filter out false Greek signals.

**What's NOT Important:** Notably absent from top 5: raw net\_gamma, net\_vanna, net\_dgex (the features traders typically watch). The model learned that derivatives and positioning matter more than headline numbers.

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## Usage Notes

### Feature Scaling

- **Distance features:** Already normalized by dividing by spot\_price (scale ~0.001 to 0.05)
- **Greek exposures:** Raw values (scale varies by ticker, model learns appropriate weighting)
- **Binary flags:** {0, 1}
- **Normalized features:** [0, 1] range (IV percentile, RSI, time features)
- **Ratios:** Can be unbounded; model handles naturally

# Missing Data Handling

- **Missing Greek data:** Use zeros (indicates no options flow that period)
- **Missing IB data:** Skip that day entirely (IB is critical framework)
- **Missing IV data:** Use 0.5 for percentile, 0 for zscore (neutral assumption)
- **Missing VIX data:** Use regime=1 (elevated/neutral assumption)

## Computational Notes

- **Greek files:** ~1-2 per minute during market hours (9:30-16:00 ET)
  - **Processing time:** ~12 minutes for 14 days × 5 tickers (parallelized)
  - **Dataset size:** 93,387 samples = ~14 days × 5 tickers × ~1340 snapshots/day
  - **Feature count:** 74 numerical features + 5 meta features
- 

## References

### Data Sources

- **Greek Exposures:** SpotGamma-style calculations from options chains
- **IB Methodology:** Dalton's Market Profile framework
- **Fibonacci Ratios:** Classical technical analysis (1.272, 1.618, 2.0)
- **VIX:** CBOE Volatility Index (ticker: VIX)

### Key Concepts

- **Gamma Pinning:** Price gravitates toward strikes with high positive gamma
- **Charm:**  $d\text{Gamma}/d\text{Time}$  - how gamma changes with time decay
- **Vanna:**  $d\text{Delta}/d\text{Vol}$  - how directional exposure changes with volatility
- **Zomma:**  $d\text{Gamma}/d\text{Vol}$  - how gamma changes with volatility
- **Vega:**  $d\text{Price}/d\text{Vol}$  - sensitivity to volatility changes
- **Vomma:**  $d\text{Vega}/d\text{Vol}$  - convexity of vega (second order)

## Feature Reduction Strategy

The model uses a 3-step feature reduction pipeline ( `neural/feature_reduction.py` ) to manage the large number of features (~93):

1. **Variance Filtering:** Removes features with near-zero variance ( $< 0.01$ ) to eliminate constant or irrelevant signals.

2. **Correlation Grouping:** Identifies groups of highly correlated features ( $|r| > 0.85$ ).
3. **Grouped PCA:** Applies PCA to each correlated group (keeping top 3 components) while preserving standalone features.

This reduces dimensionality while retaining the signal from correlated groups (e.g., if multiple Greek distances are highly correlated, PCA extracts their common trend).

## Model Architecture

- **Type:** Hybrid LSTM + Dense Neural Network
  - **Framework:** PyTorch with CUDA acceleration
  - **Size:** Medium (128 hidden, 256→128→64 MLP)
  - **Training:** 200 epochs, best validation: 88.7% accuracy
  - **Generalization:** Excellent train-val gap
- 

## Changelog

**v1.0** - Initial feature set with ODTE Greeks, IB features, basic IV

**v2.0** - Added weekly Greeks, cross-expiry divergence, VIX features

**v3.0** - Added Fourier IV processing, engineered features (ratios, deltas, cross-features)

**v3.1** - Added DGEX key levels and price\_vs\_dgex\_magnet feature

**v3.2 - Major discovery: net\_charm is dominant feature (21.5%)** - theta decay pressure is primary edge

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*Last Updated: 2026-02-15*

*Model Version: trading\_hybrid.pt (medium)*

*Training Dataset: 93,387 samples, 14 days, 88.2% final accuracy*

*Top Feature: net\_charm (21.5% importance) - Time decay directional pressure*